

Jonathan A Diller

Lakewood, CO | Jonathan.A.Diller@Gmail.com | Updated: Mar. 2025

Interests and Goals

My research is in systems-aware planning and tasking for multi-robot teams. A common theme in my work is incorporating realistic models of the world into planning algorithms, using both centralized and decentralized approaches. I also have a general interest in AI and Math Programming. My career goal is to remain in academia and start my own research lab.

Education

Colorado School of Mines	Golden, CO
<i>Doctor of Philosophy in Robotics (In Progress)</i>	<i>Expected Graduation: 2025</i>

Colorado School of Mines	Golden, CO
<i>Master of Science in Computer Science</i>	May 2022
GPA: 4.0/4.0	

Pennsylvania State University, Harrisburg	Middletown, PA
<i>Bachelor of Science in Computer Science</i>	May 2020
- Minor in Mechatronics Technology - Minor in Mathematics - Graduated summa cum laude - GPA: 3.99/4.0	

Research Experience

Research Assistant (<i>Under Dr. Qi Han</i>)	July 2020 - Present
<i>Pervasive Computing Systems Group, Colorado School of Mines</i>	<i>Golden, CO</i>
Studying and implementing multi-robot projects with focus on communication and energy aware planning.	

Journeyman Fellow (<i>Mentor: Dr. J. Humann and J. Dotterweich</i>)	June 2024 - August 2024
<i>Army Research Laboratory</i>	<i>Golden, CO (remote)</i>
Researched and developed planning algorithms for mixed drone-UGV teams.	

Graduate Research Associate (<i>Mentor: Dr. John Rogers</i>)	May 2023 - August 2023
<i>Army Research Laboratory</i>	<i>Adelphi, MD</i>
Researched the topics of task allocation and communication mapping for robot teams.	

Research Scholar (<i>Mentor: Dr. Peter Idowu</i>)	June 2019 - July 2019
<i>Penn State Drawdown REU Program</i>	<i>Middletown, PA</i>
Designed and evaluated algorithms for controlling microgrids using PLCs.	

Research Assistant (<i>Under Dr. Javad Khazaei</i>)	March 2019 - June 2019
<i>Pennsylvania State University</i>	<i>Middletown, PA</i>
Researched and developed microcontroller applications for use in renewable energy projects.	

Publications

- 2025** J. Diller, Q. Han, R. Byers, J. Dotterweich, and J. Humann. "*Hitchhiker's Guide to Patrolling: Path-Finding for Energy-Sharing Drone-UGV Teams.*" Accepted to the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2025.
- 2025** J. Diller, and Q. Han. "*Energy-Aware Drone Path Finding with a Fixed-Trajectory Ground Vehicle.*" ACM Journal on Autonomous Transportation Systems.
- 2024** C. Wang, J. Diller, and Q. Han. "*LLM for Generating Simulation Inputs to Evaluate Path Planning Algorithms.*" In Proceedings of the International Conference on Machine Learning and Applications (ICMLA), 2024.
- 2023** J. Diller, P. Hall, Q. Han. "*Holistic path planning for multi-drone data collection.*" In Proceedings of the 19th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), 2023
- 2023** J. Diller, N. Dantam, J. Rogers, Q. Han. "*Communication Jamming-Aware Robot Path Adaptation.*" In Proceedings of the Second Workshop on Distributed Collective Intelligence (DISCOLI), 2023
- 2023** P. Hall, J. Diller, A. Moon, Q. Han. "*DroNS-3: Framework for Realistic Drone and Networking Simulators.*" In Proceedings of the Ninth Workshop on Micro Aerial Vehicle Networks, Systems, and Applications (DroNet), 2023
- 2023** J. Diller and Q. Han. "*Energy-aware UAV Path Planning with Adaptive Speed.*" In Proceedings of the 22nd International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2023
- 2022** J. Diller, P. Hall, C. Schanker, K. Ung, P. Belous, P. Russell, and Q. Han. "*ICCSwarm: A Framework for Integrated Communication and Control in UAV Swarms.*" In Proceedings of the Eighth Workshop on Micro Aerial Vehicle Networks, Systems, and Applications (DroNet), 2022
- 2020** J. Diller, P. Idowu, J. Khazaei, "Load-Leveling Trainer for Demand Side Management on a 45kW Cyber-Physical Microgrid," Texas Power and Energy Conference 2020 (TPEC), 2020
- 2020** J. Diller, B. Trussell, J. Khazaei, P. Idowu, "Hardware Development of a Three-Phase 3.5 kW SiC Converter with Sinusoidal PWM," Texas Power and Energy Conference 2020 (TPEC), 2020

Awards & Recognitions

- 2024 Cyber-Physical Systems Rising Start **May, 2024**
- Best PhD Poster, C-MAPP, Colorado School of Mines **Jan, 2024**
- Best PhD Poster, C-MAPP, Colorado School of Mines **Jan, 2023**

- Best Elevator Pitch, C-MAPP, *Colorado School of Mines* Feb, 2022
- Graduation Student Marshal for School of Sci., Eng. and Tech. May, 2020
- Computer Science Outstanding Student, *Pennsylvania State University* Apr, 2020
- Evan Pugh Scholar Award - Senior, *Pennsylvania State University* Apr, 2019
- Evan Pugh Scholar Award - Junior, *Pennsylvania State University* Apr, 2018
- President's Freshman Award, *Pennsylvania State University* Apr, 2017
- Dean's List, *Pennsylvania State University* Spring 2016 - Spring 2020

Teaching Experience

Primary Instructor August 2024 – Present
Colorado School of Mines Golden, CO

- CSCI 128: Computer Science for STEM (Fall 2024, Spring 2025)

Teaching Assistant August 2020 – December 2022
Colorado School of Mines Golden, CO

- CSCI 565: Distributed Systems (Fall 2022)
- CSCI 406: Algorithms (Spr 2021)
- CSCI 261: Programming Concepts (Fall 2020)

Peer Tutor August 2017 – May 2020
Pennsylvania State University Middletown, PA

- Tutor students in Computer Science, Mathematics, Physics and Engineering classes.
- Earned CRLA Certified Tutor, Level I Certification.

Work Experience

Robert Bosch GmbH May 2018 – July 2020
Embedded Software Developer & Test Intern Lancaster, PA

- Develop firmware for embedded real-time environments.
- New product prototyping.
- Design and implement automated tests for moving cameras.
- Write documentation for internal procedures.

United States Marine Corps November 2010 - November 2015
KC-130J Crewmaster, Plane Captain San Diego, CA & Okinawa, Japan

- Last held rank/pay grade: Sergeant/E-5
- Supervised small teams in pre and post flight inspections on C-130J aircraft.

Technical Skills

- Programming Languages: C/C++, Python, Java, Matlab, Scheme
- Real-Time and Embedded Software Development
- ROS/ROS2
- Gurobi Optimizer
- Software Development in Windows and Linux Environments